

PROJECT REPORT:- **LIBRARY MANAGEMENT**

Subject: Introduction of Problem Solving

Program: BTech Computer Science and
Engineering (SCAI)

Semester: Fall Semester 2026-27

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INTRODUCTION

A library contains many books, and a librarian needs to know which books are available, which books have been issued, and which books have been returned. When these records are maintained only on paper, searching and updating information can take time. The Library Management System is a menu-driven Python program designed to make these basic tasks easier. The user can add book details, display all books, search for a book, issue an available book to a student, return a book, calculate a late-return fine, and view a summary report. The project is designed for a first-semester student. It uses simple Python concepts such as variables, input and output, conditional statements, loops, functions, lists, dictionaries, and basic validation. It does not require an external package or database

PROJECT STATEMENT :-

Maintaining library records manually can lead to slow searches, duplicate entries, and confusion about whether a book is available or already issued. It can also be difficult to count available books or calculate a late-return fine consistently. The problem addressed by this project is to create a simple computer-based system that stores book details during a session and lets the user perform common library operations through a clear menu. The program should check book availability before issuing a book and update the book status when it is returned.

OBJECTIVE :-

The main objectives of this project are:

1. To maintain a record of books in the library.
2. To add and display book information.
3. To search for a particular book.
4. To issue books to students.
5. To return issued books.
6. To calculate fines for late returns.
7. To generate a simple library report.
8. To practice Python programming concepts learned in the first semester.

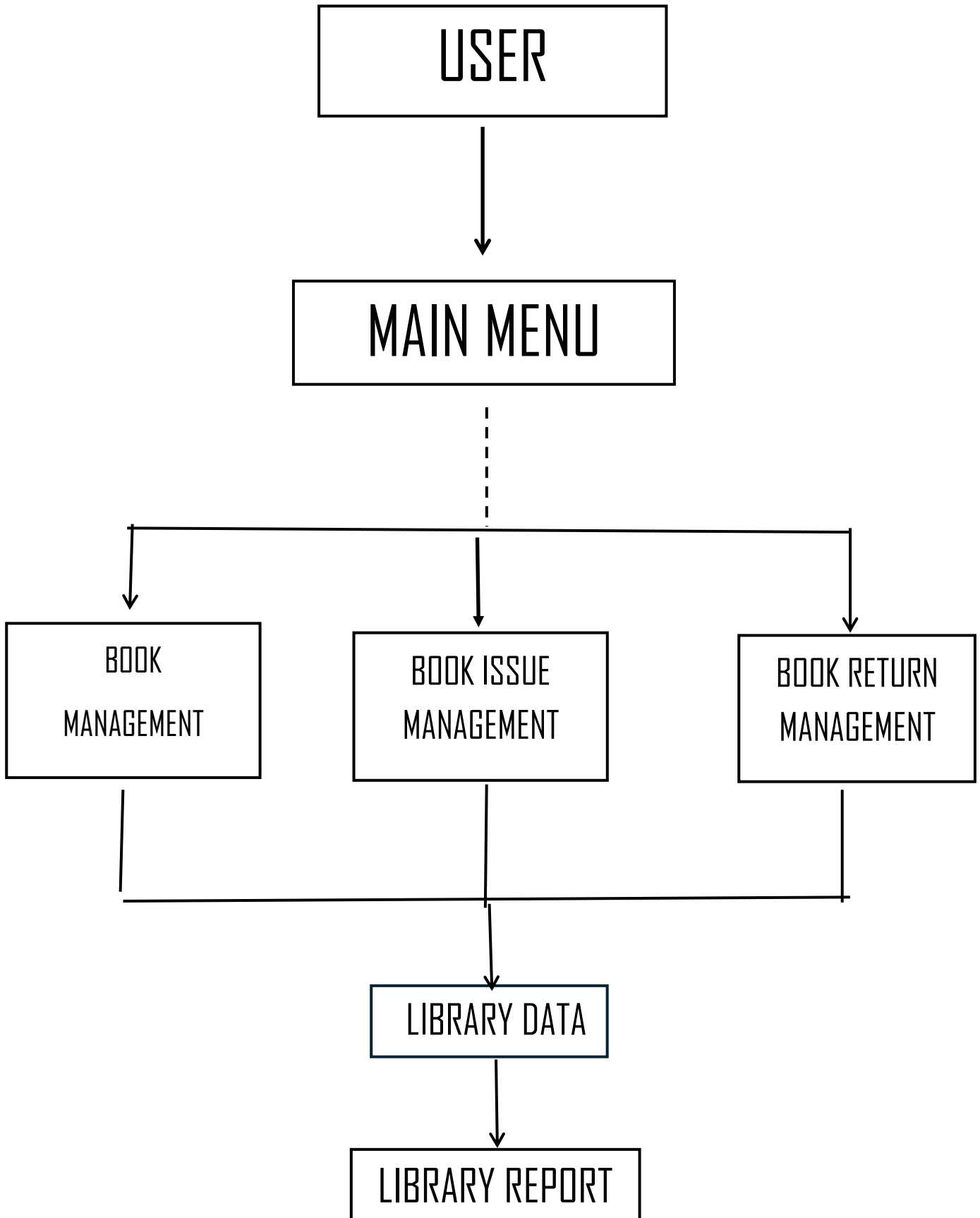
FUNCTIONAL REQUIERMENTS :-

1. **Add Book**:- Allows the librarian to add a new book.
2. **Display Books**:- Displays all books and their availability.
3. **Search Book**:- Searches for a book using its ID or title.
4. **Issue Book**:- Issues an available book to a student.
5. **Return Book**:-Returns an issued book and updates its status.
6. **Calculate Fine**:- Calculates the fine if a book is returned late.
7. **Library Report**:- Displays total books, available books, issued books, and total fine.

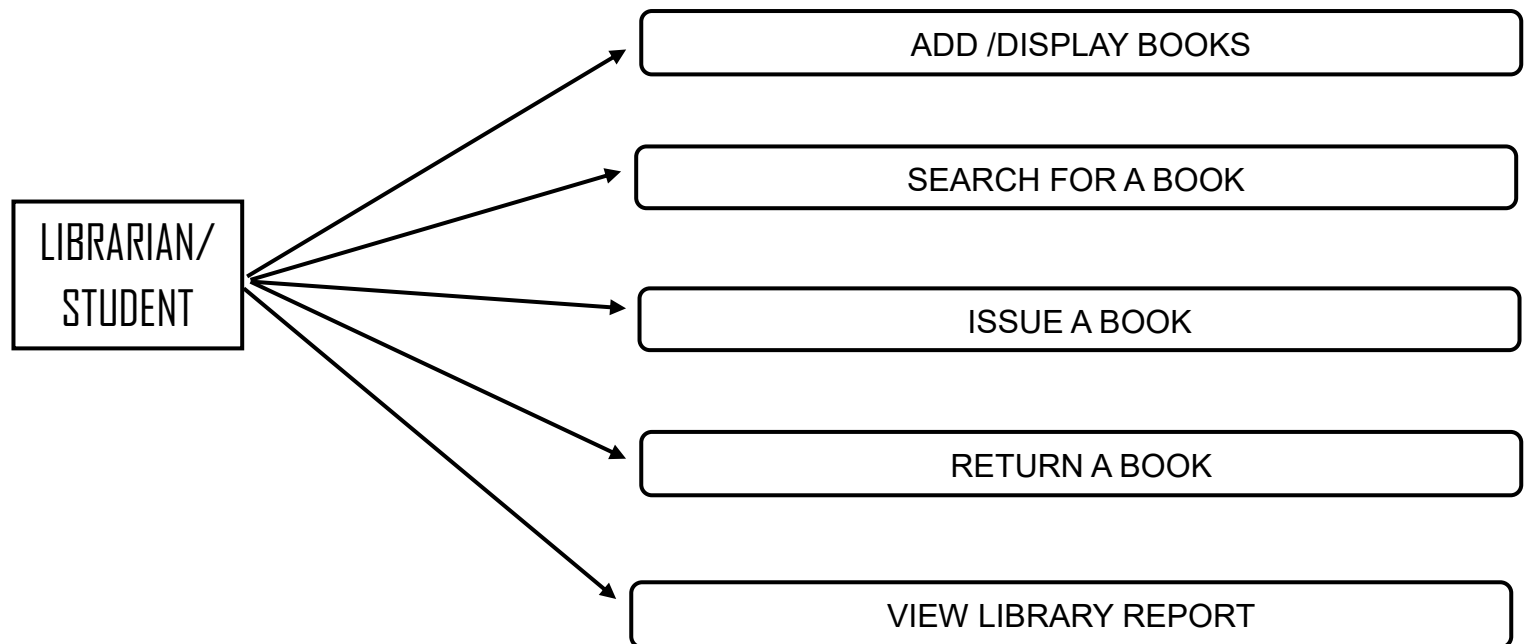
NON FUNCTIONAL REQUIERMENTS :-

- 1. Usability :-**The program should have a simple menu so that users can easily understand and use the system.
- 2. Reliability :-**The program should give correct results for adding, searching, issuing, and returning books.
- 3. Error Handling :-**The program should handle invalid menu choices and should not allow an already issued book to be issued again.
- 4. Maintainability :-**The program is divided into separate functions, making it easier to understand, modify, and maintain.
- 5. Performance :-**The program should provide results quickly for normal library operations.
- 6. Security :-**Only authorized users such as the librarian should be allowed to perform management operations.

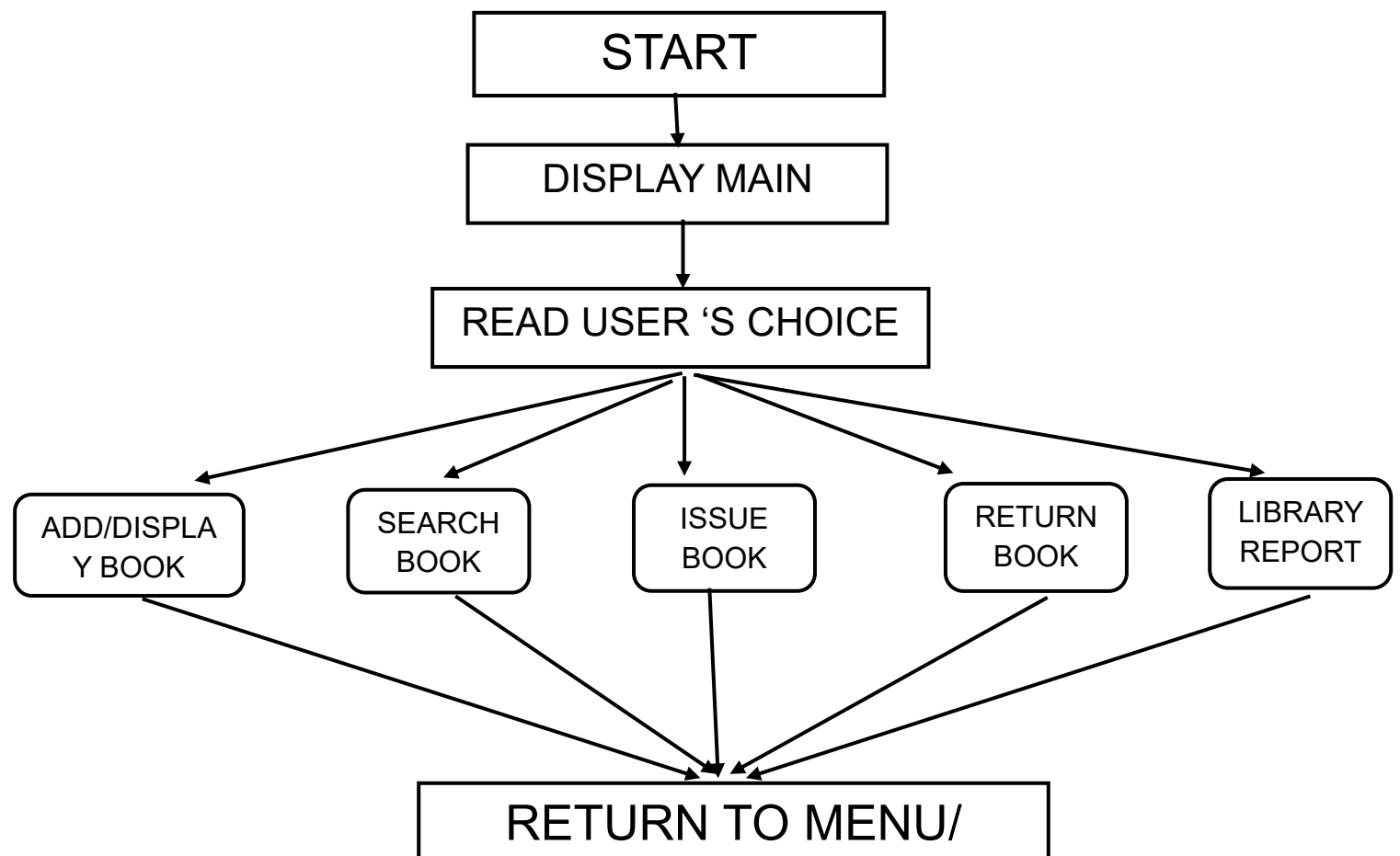
SYSTEM ARCHITECTURE



USE CASE DIAGRAM :-

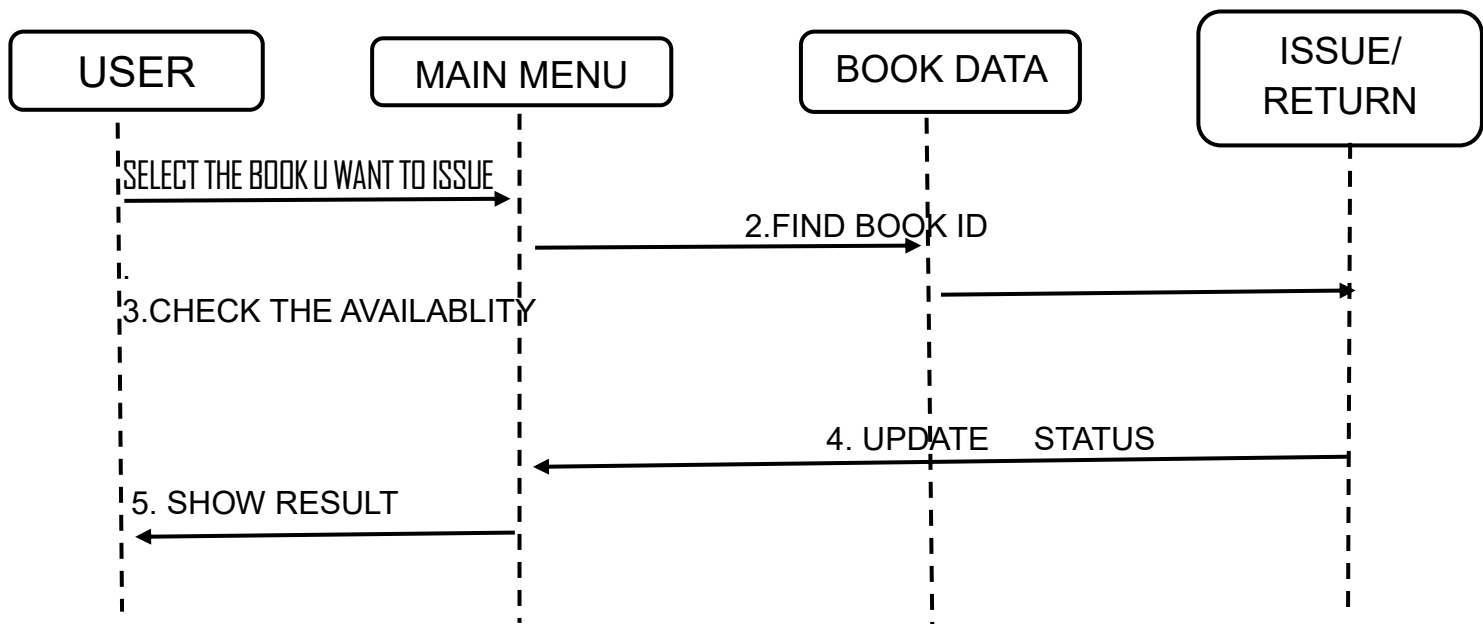


WORKFLOW DIAGRAM :-

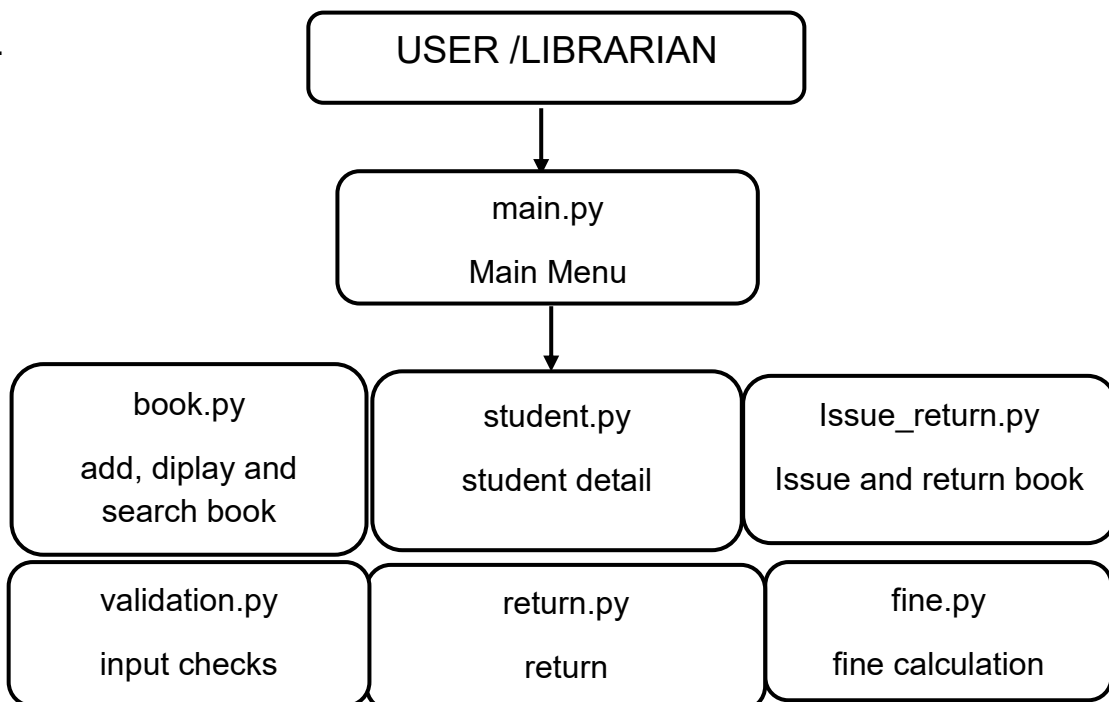


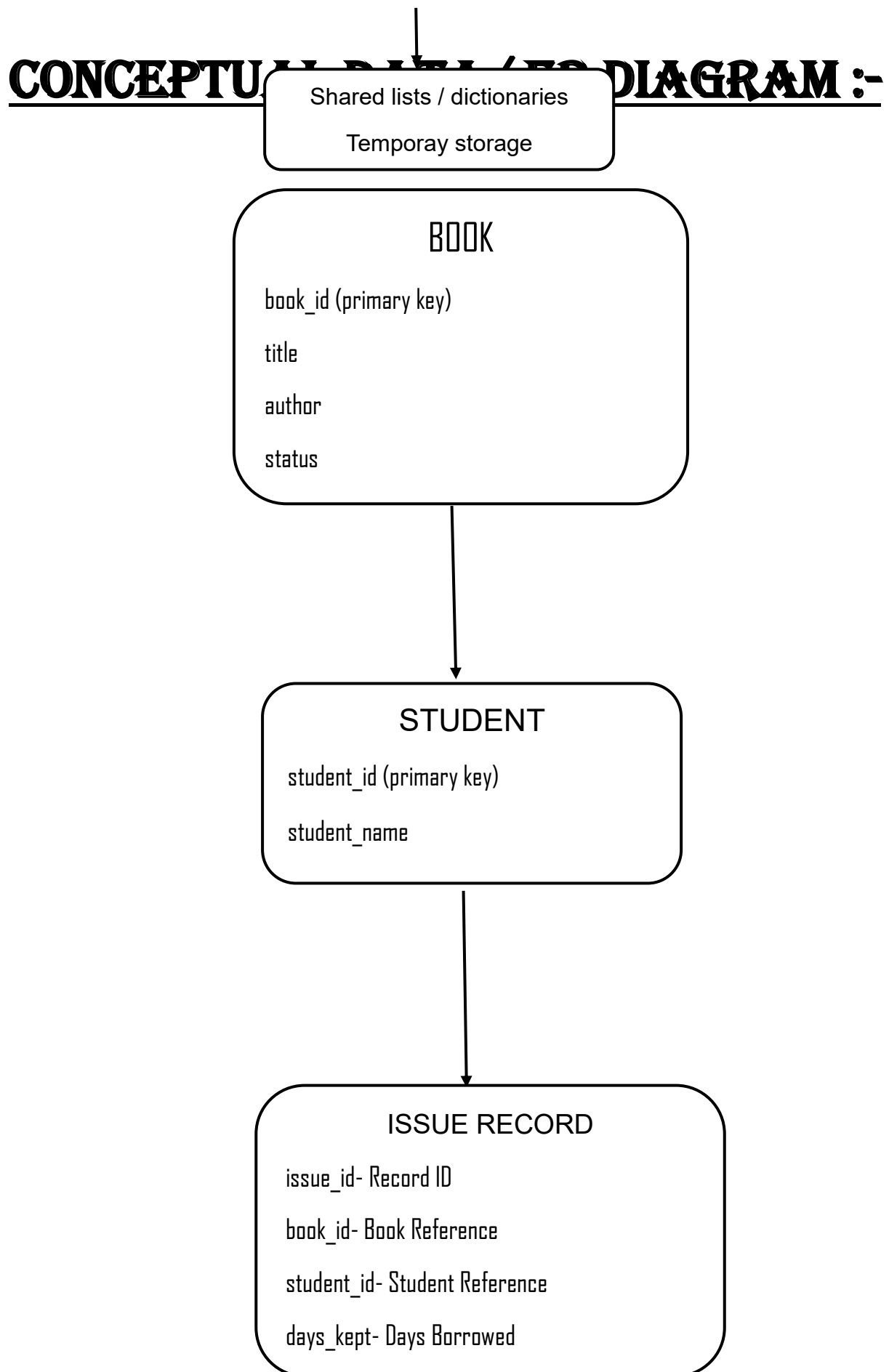
SEQUENCE DIAGRAM : ISSUING A BOOK:-

This sequence shows how an issue request moves between the user, main menu, book data, and issue/return logic



COMPONENT DIAGRAM :-





DESIGN DECISION & RATIONALE :

- **I used a command-line interface** :- It is easy to build and understand in a first-semester course and keeps attention on Python logic.
- **I used Book ID as the main identifier** :- An ID is more reliable for selecting a book than its title, which may be shared by different books.
- **I used a simple fixed fine rule**:- The calculation is easy to explain and verify: Rs 5 per day after the 7-day allowance. (Rationale for the fine rule For a book kept for 7 days or fewer, the fine is zero. For a book kept longer, the number of late days is calculated as (days kept - 7), and the fine is late days $\times$ 5. For example, 10 days results in 3 late days and a fine of Rs 15. The fine policy is a sample rule for demonstration)

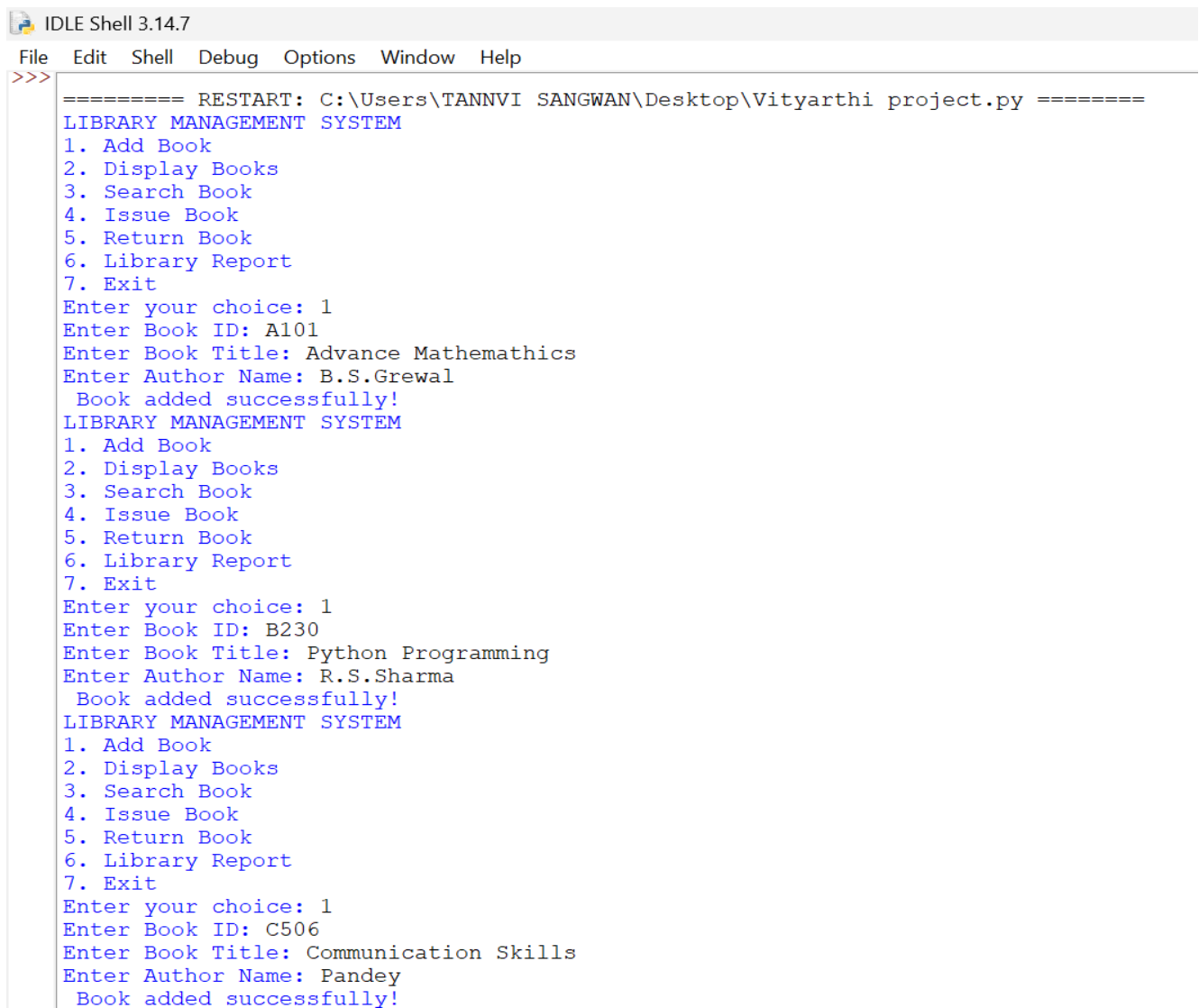
IMPLEMENTATION DETAILS:

- The books list stores one dictionary per book. A dictionary can contain the book ID, title, author, and status. The issued_books list stores the book ID and the student's name and ID for each issue record
- Functions :- Each operation has a clear purpose and can be tested separately.

- Lists and Dictionaries :- These built-in structures are enough to store several book and issue records without introducing a database.
- Menu loop :- The user can perform more than one operation in a single run.

SCREENSHOTS:-

1. ADDING BOOKS:-



```

IDLE Shell 3.14.7
File Edit Shell Debug Options Window Help
>>> ===== RESTART: C:\Users\TANNVI SANGWAN\Desktop\Vityarthi project.py =====
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 1
Enter Book ID: A101
Enter Book Title: Advance Mathematics
Enter Author Name: B.S.Grewal
  Book added successfully!
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 1
Enter Book ID: B230
Enter Book Title: Python Programming
Enter Author Name: R.S.Sharma
  Book added successfully!
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 1
Enter Book ID: C506
Enter Book Title: Communication Skills
Enter Author Name: Pandey
  Book added successfully!

```

2. DISPLAYING BOOKS:-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 2
BOOK LIST
Book ID : A101
Title   : Advance Mathematics
Author  : B.S.Grewal
Status  : Available
Book ID : B230
Title   : Python Programming
Author  : R.S.Sharma
Status  : Available
Book ID : C506
Title   : Communication Skills
Author  : Pandey
Status  : Available
```

3. SEARCHING BOOKS :-

```
Status : Available
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 3
Enter Book ID or Title to search: A101
Book Found :)
Book ID : A101
Title   : Advance Mathematics
Author  : B.S.Grewal
Status  : Available
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 3
Enter Book ID or Title to search: A202
Book not found.
```

4. ISSUING BOOKS:-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 4
Enter Book ID to issue: A101
Enter Student Name: Tannvi
Enter Student ID: 2001
Book issued successfully!
Student Name: Tannvi
Book: Advance Mathematics
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 4
Enter Book ID to issue: C506
Enter Student Name: Shubhra
Enter Student ID: 1001
Book issued successfully!
Student Name: Shubhra
Book: Communication Skills
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 4
Enter Book ID to issue: A101
Enter Student Name: Vidhya
Enter Student ID: 4002
Book is already issued.
```

5. LIBRARY REPORT:-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 6
LIBRARY REPORT
Total Books      : 3
Available Books  : 1
Issued Books     : 2
```

6. RETURNING BOOK:-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 5
Enter Book ID to return: A101
Enter number of days book was kept: 6
Book returned successfully!
No fine.
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 5
Enter Book ID to return: C506
Enter number of days book was kept: 12
Book returned successfully!
Fine to be paid: ₹ 25
```

7. CHOSING WRONG OPTION:-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 8
  Invalid choice :(
-----
```

8.EXITING THE PROGRAM :-

```
LIBRARY MANAGEMENT SYSTEM
1. Add Book
2. Display Books
3. Search Book
4. Issue Book
5. Return Book
6. Library Report
7. Exit
Enter your choice: 7
  Thanks for using the Library Management System <3
```

TESTING APPROACH :-

- Add a book with a new ID Book record is added and success message is shown.
- Display books when the list is empty Program shows a clear "No books available" message.
- Search using an existing Book ID Correct book details are displayed.
- Search using a missing Book ID Program displays "Book not found".
- Issue a book marked Available Status changes to Issued and student details are recorded.
- Issue the same book a second time Program refuses the request because the book is already issued
- Return an issued book kept for 7 days Book becomes Available and fine is 0.
- Return an issued book kept for 10 days Fine is calculated as $(10 - 7) \times n5 = n15$.
- Enter an invalid menu choice Program displays an error message and returns to the menu
- Choose Exit Menu loop ends

CHALLENGES FACED :-

1. Keeping book status correct: the program must not issue a book that is already issued. This is handled by checking the status before changing it.
2. Matching an issue to the correct book: using Book ID makes it easier to find the intended record and update its status.
3. Updating records across operations: issue and return functions must use the same book data so that the display and report show current information.
4. Handling incorrect input: a user may enter a missing ID, an invalid menu option, or letters where a number is expected. Input validation is needed to prevent unexpected errors.
5. Organising multiple files: when the program is split into modules, each file must contain the right functions and imports. The main file should coordinate the modules without duplicating logic.
6. Temporary storage limitation: lists and dictionaries lose their contents when the program closes. This is acceptable for the first prototype but should be improved if the system is extended

LEARNING AND KEY TAKEAWAY :-

- Writing functions to divide a large program into smaller tasks
- Using lists to hold multiple records.
- Using dictionaries to represent the details of one book or student.
- Using loops to display records and keep the menu active.
- Using conditional statements to make decisions about book availability.
- Taking input from a user and displaying useful messages.
- Designing a simple algorithm for fine calculation.
- Testing normal and invalid cases.
- Organising code into separate files for maintainability.

FUTURE ENHANCEMENTS :

1. Better student records: allow the user to register, search, and update student details.
2. Due-date support: File handling: save book and issue records in a text, CSV, or JSON file so data remains after the program closes.
3. Login system: add a librarian login and limit access to management operations
4. Use dates instead of asking the user to enter the total number of days
5. Book update and removal: allow the librarian to edit book information or remove a book that is not issued.
6. Improved reports: show overdue books, most-issued books, and total fines by student.
7. Graphical interface: build a simple GUI after learning a suitable Python library

REFERENCE:-

- VITyarthi course : Python Essentials by D.LAXMI